Course Notes

Applied Abstract Algebra
Quiver Theory

Instructor:

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TA:

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Contents

Chapter 1

Lecture 1

Course Information

Main topic: Representation theory of quivers.

Office hours: Mon: 12:30-14:30.

Main book: *Ralf Schiffler, Quiver Representations*, Cambridge University Press.

References:

- *Assem, Simons, Skowronski, Elements of the Representation Theory of Associative Algebras*, Cambridge University Press.
- *Ibrahim Assem, basic representation theory of algebras.*
- *Michel Brion, Representations of quivers.*
- *Henning Krause, Representations of quivers via reflection functor.*
- *W.Crawley-Boevey, Lectures on representations of quivers.*
- *W.Crawley-Boevey, More on representations of quivers.*
- *Alexander Kirillov Jr. - Quiver Representations and Quiver Varieties (2016, American Mathematical Society).*
- Other Reference: `www.math.uni-bielefeld.de`

Grading:

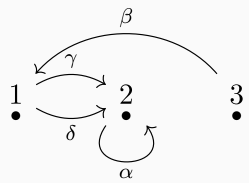
- Midterm: 30%
- Presentation: 10%
- Homework: 20%
- Final: 40%

Midterm will be on April 13th, Monday, 2026 (the 8th week).

1.1 Quivers

Definition 1.1.1. *A quiver Q is a quadruple (Q_0, Q_1, s, t), where Q_0 is a set of vertices, Q_1 is a set of arrows, and $s, t : Q_1 \to Q_0$ are two maps. For any arrow $\alpha \in Q_1$, $s(\alpha)$ is called the source of α and $t(\alpha)$ is called the target of α. For example, if $\alpha : 1 \to 2$, then $s(\alpha) = 1$ and $t(\alpha) = 2$.*

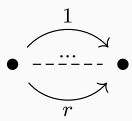
Example 1.1.2. *Let* $Q_0 = \{1, 2, 3\}$*,* $Q_1 = \{\alpha, \beta, \gamma, \delta\}$*. We have* $s(\alpha) = 2, t(\alpha) = 2$*;* $s(\beta) = 3, t(\beta) = 1$*;* $s(\gamma) = s(\delta) = 1, t(\gamma) = t(\delta) = 2$*.*



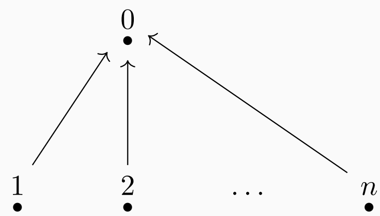
Let path $\xi = \alpha\gamma\beta$*, then we have* $s(\xi) = s(\beta) = 3$ *and* $t(\xi) = t(\alpha) = 2$*.*

Example 1.1.3. *Jordan quiver. The Jordan quiver consists of a single vertex and a single arrow starting and ending at this vertex. Formally,* $Q_0 = \{1\}$ *and* $Q_1 = \{\alpha\}$ *with* $s(\alpha) = t(\alpha) = 1$*. It is called the Jordan quiver because its representations correspond to vector spaces equipped with a linear endomorphism. The classification of such representations up to isomorphism is equivalent to the classification of square matrices up to similarity, which is given by the Jordan normal form theorem (over an algebraically closed field).*

Example 1.1.4. *r-Kronecker quiver. The r-Kronecker quiver has two vertices, say 1 and 2, and r arrows* $\alpha_1, \ldots, \alpha_r$ *from 1 to 2. Thus* $Q_0 = \{1, 2\}$*,* $Q_1 = \{\alpha_1, \ldots, \alpha_r\}$*, with* $s(\alpha_i) = 1$ *and* $t(\alpha_i) = 2$ *for all* $i = 1, \ldots, r$*. It is named after Leopold Kronecker. In the case* $r = 2$*, the representations correspond to pairs of linear maps between two vector spaces, which can be represented by matrix pencils. Kronecker solved the classification problem for these pencils.*



Example 1.1.5. *n-subspace quiver. The n-subspace quiver consists of* $n + 1$ *vertices* $\{0, 1, \ldots, n\}$ *and n arrows* $\alpha_1, \ldots, \alpha_n$ *such that* $s(\alpha_i) = i$ *and* $t(\alpha_i) = 0$ *for all* $i = 1, \ldots, n$*. It is called the n-subspace quiver because its representations are related to configurations of subspaces. If the linear maps associated with the arrows are injective, a representation corresponds to a collection of n subspaces of the vector space associated with the central vertex.*



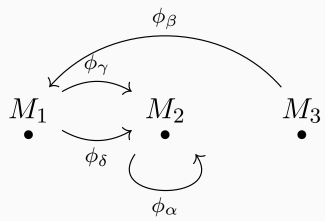
Example 1.1.6. *Finite quivers. A quiver $Q = (Q_0, Q_1, s, t)$ is called finite if both the set of vertices Q_0 and the set of arrows Q_1 are finite sets.*

For the base field we assume $k = \bar{k}$. Usually we take $k = \mathbb{C}$.

1.2 Representations

Definition 1.2.1. *A representation M of a quiver Q consists of a vector space M_i for each vertex $i \in Q_0$ and a linear map $\phi_\alpha : M_{s(\alpha)} \to M_{t(\alpha)}$ for each arrow $\alpha \in Q_1$. We denote this representation by $M = (M_i, \phi_\alpha)_{i \in Q_0, \alpha \in Q_1}$.*

Example 1.2.2. *For example 1:*



Definition 1.2.3. *A representation M is called **finite dimensional** if $\dim M_i < \infty$ for all $i \in Q_0$. The tuple $\dim M = \{\dim M_i\}_{i \in Q_0}$ is called the **dimension vector** of M. An element $m \in M$ is a tuple $\{m_i\}_{i \in Q_0}$ where $m_i \in M_i$ for each $i \in Q_0$.*

Example 1.2.4. 1-*Kronecker quiver:*

$$1 \longrightarrow 2$$

Take k to be a field. Some representations of this quiver are:

$$M: \quad k \xrightarrow{\;1\;} k \qquad M': \quad k \xrightarrow{\;0\;} k \qquad M'': \quad k \xrightarrow{\;0\;} 0$$

Another example is

$$\widetilde{M}: \quad k^2 \xrightarrow{\begin{bmatrix} 1 & 0 \\ 1 & 0 \\ 0 & 0 \end{bmatrix}} k^3 \ .$$

For $v = (1, 0) \in k^2$, we have

$$\begin{bmatrix} 1 & 0 \\ 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}.$$

1.3 Morphisms and the Category $\operatorname{Rep} Q$

Definition 1.3.1 (Morphism between two representations of Q)**.** *Let $M = (M_i, \phi_\alpha)_{i \in Q_0, \alpha \in Q_1}$ and $N = (N_i, \psi_\alpha)_{i \in Q_0, \alpha \in Q_1}$ be two representations of Q. A morphism $f : M \to N$ is a collection of linear maps $f_i : M_i \to N_i$ such that for every $\phi_\alpha : M_i \to M_j$, $\alpha \in Q_1$ the following diagram commutes:*

$$\begin{tikzcd} M_i \arrow[r, "\phi_\alpha"] \arrow[d, "f_i"'] & M_j \arrow[d, "f_j"] \\ N_i \arrow[r, "\psi_\alpha"'] & N_j \end{tikzcd}$$

Example 1.3.2. *In example of 1-Kronecker quiver, let $f : M \to M''$, $f_1 : k \to k$ such that $f_1(1) = a$, and $f_2 : k \to 0$ is the zero map. We can check that f is a morphism. Check the following diagram:*

$$\begin{tikzcd} & k \arrow[r, "1"] \arrow[d, "a"'] & k \arrow[d, "0"] & \\ f_1 & k \arrow[r, "0"'] & 0 & f_2 \end{tikzcd}$$

Now we establish $g : M'' \to M$. If g is a morphism, then we have $g_1 : k \to k$ such that $g_1(1) = b$, and $g_2 : 0 \to k$ is the zero map. To make the target diagram commute it forces $g_1 = 0$. Hence there are only 0 morphism form M'' to M.

$$\begin{tikzcd} & k \arrow[r, "0"] \arrow[d, "b"'] & 0 \arrow[d, "0"] & \\ g_1 & k \arrow[r, "1"'] & 0 & g_2 \end{tikzcd}$$

$$1 \circ g_1 = 0 \Rightarrow g_1 = 0$$

Recall the definition of category:

Definition 1.3.3. *A category $\mathcal{C}$ consists of the following data:*

- *A class of objects, denoted by $\mathrm{Ob}(\mathcal{C})$.*
- *For any pair of objects $X, Y \in \mathrm{Ob}(\mathcal{C})$, a set of morphisms from X to Y, denoted by $\mathrm{Hom}_{\mathcal{C}}(X, Y)$. If $f \in \mathrm{Hom}_{\mathcal{C}}(X, Y)$, we write $f : X \to Y$.*
- *For any three objects $X, Y, Z \in \mathrm{Ob}(\mathcal{C})$, a composition map:*

$$\circ : \mathrm{Hom}_{\mathcal{C}}(Y, Z) \times \mathrm{Hom}_{\mathcal{C}}(X, Y) \to \mathrm{Hom}_{\mathcal{C}}(X, Z), \quad (g, f) \mapsto g \circ f.$$

These data must satisfy the following axioms:

1. ***Associativity**: For any morphisms $f : X \to Y$, $g : Y \to Z$, and $h : Z \to W$,*

we have

$$h \circ (g \circ f) = (h \circ g) \circ f.$$

2. ***Identity:*** *For every object* $X \in \mathrm{Ob}(\mathcal{C})$*, there exists an identity morphism* $\mathrm{id}_X \in \mathrm{Hom}_{\mathcal{C}}(X, X)$ *such that for any* $f : X \to Y$ *and* $g : Z \to X$*,*

$$f \circ \mathrm{id}_X = f \quad \textit{and} \quad \mathrm{id}_X \circ g = g.$$

Denote Rep$Q$ the category of representations of Q. Then Ob(Rep$Q$) is the class of representations of Q: $M = (M_i, \phi_\alpha)$, and $\mathrm{Hom}_{\mathrm{Rep}Q}$ is the class of morphisms $\mathrm{Hom}_{\mathrm{Rep}Q}(M, N)$ from M to N.

Let $M, N \in \mathrm{Ob}(\mathrm{Rep}Q)$. Then $\mathrm{Hom}(M, N) := \mathrm{Hom}_{\mathrm{Rep}Q}(M, N)$ is a vector space over k with operations defined component-wise: for $f, g \in \mathrm{Hom}(M, N)$ and $\lambda \in k$, let $(f + g)_i = f_i + g_i$ and $(\lambda f)_i = \lambda f_i$ for all $i \in Q_0$.

Verification: Let $M = (M_i, \phi_\alpha)$ and $N = (N_i, \psi_\alpha)$. Since f, g are morphisms, for any arrow $\alpha : i \to j$, we have $\psi_\alpha f_i = f_j \phi_\alpha$ and $\psi_\alpha g_i = g_j \phi_\alpha$.

1. **Addition:**

$$\psi_\alpha (f + g)_i = \psi_\alpha (f_i + g_i) = \psi_\alpha f_i + \psi_\alpha g_i = f_j \phi_\alpha + g_j \phi_\alpha = (f_j + g_j)\phi_\alpha = (f + g)_j \phi_\alpha.$$

Thus $f + g \in \mathrm{Hom}(M, N)$.

2. **Scalar Multiplication:**

$$\psi_\alpha (\lambda f)_i = \psi_\alpha (\lambda f_i) = \lambda(\psi_\alpha f_i) = \lambda(f_j \phi_\alpha) = (\lambda f_j)\phi_\alpha = (\lambda f)_j \phi_\alpha.$$

Thus $\lambda f \in \mathrm{Hom}(M, N)$.

The zero morphism $0 = (0)_{i \in Q_0}$ clearly satisfies the commutativity condition. The vector space axioms follow from the structure of linear maps between vector spaces.

Example 1.3.4. *Consider*

$$1 \longrightarrow 2$$

$\mathrm{Hom}(M, M'') = \{(a, 0) | a \in k\} \simeq k$. $\mathrm{Hom}(M'', M) = \{0\}$.

Example 1.3.5. *Let* Q *be a quiver:*

$$\begin{tikzcd}
1 \arrow[dr, "\alpha"] & & \\
& 3 & 4 \arrow[l, "\gamma"'] \\
2 \arrow[ur, "\beta"] & &
\end{tikzcd}$$

Let M*:*

$$
\begin{tikzcd}
k \arrow[dr, "\begin{bmatrix}1\\0\end{bmatrix}"] & & \\
& k^2 & k \arrow[l, "\begin{bmatrix}1\\1\end{bmatrix}"'] \\
k \arrow[ur, "\begin{bmatrix}0\\1\end{bmatrix}"] & &
\end{tikzcd}
$$

M':

$$
\begin{tikzcd}
0 \arrow[dr, "0"] & & \\
& k & 0 \arrow[l, "0"'] \\
0 \arrow[ur, "0"] & &
\end{tikzcd}
$$

M'':

$$
\begin{tikzcd}
k \arrow[dr, "1"] & & \\
& k & k \arrow[l, "1"'] \\
k \arrow[ur, "1"] & &
\end{tikzcd}
$$

Let us check $\operatorname{Hom}(M, M')$*:*

$$
\begin{tikzcd}
k \arrow[dr, "\begin{bmatrix}1\\0\end{bmatrix}"] \arrow[ddd, bend right, "0"'] & & \\
& k^2 \arrow[ddd, "\begin{bmatrix}a & b\end{bmatrix}"] & k \arrow[l, "\begin{bmatrix}1\\1\end{bmatrix}"'] \arrow[ddd, bend left, "0"] \\
k \arrow[ur, "\begin{bmatrix}0\\1\end{bmatrix}"] \arrow[ddd, bend right, "0"'] & & \\
0 \arrow[dr, "0"] & & \\
& k & 0 \arrow[l, "0"] \\
0 \arrow[ur, "0"] & &
\end{tikzcd}
$$

Since $M'_1 = M'_2 = M'_4 = 0$, the maps f_1, f_2, f_4 are zero. Let $f_3 = \begin{bmatrix} a & b \end{bmatrix}$. The commutativity conditions impose:

$$
\begin{cases}
f_3\phi_\alpha = \psi_\alpha f_1 \implies \begin{bmatrix} a & b \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = 0 \implies a = 0, \\
f_3\phi_\beta = \psi_\beta f_2 \implies \begin{bmatrix} a & b \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = 0 \implies b = 0.
\end{cases}
$$

Thus $f_3 = 0$*, and* $\mathrm{Hom}(M, M') = 0$*.*
Let us check $\mathrm{Hom}(M, M'')$*:*

$$\begin{tikzcd}
k \arrow[dr, "\begin{bmatrix}1\\0\end{bmatrix}"] \arrow[ddd, bend right, "c"'] & & \\
& k^2 \arrow[ddd, "{[a\ \ b]}"] & k \arrow[l, "\begin{bmatrix}1\\1\end{bmatrix}"'] \arrow[ddd, bend left, "e"] \\
k \arrow[ur, "\begin{bmatrix}0\\1\end{bmatrix}"] \arrow[ddd, bend right, "d"'] & & \\
0 \arrow[dr, "0"] & & \\
& k & 0 \arrow[l, "0"] \\
0 \arrow[ur, "0"] & &
\end{tikzcd}$$

Let the morphism be given by $g_1 = [c]$*,* $g_2 = [d]$*,* $g_4 = [e]$*, and* $g_3 = \begin{bmatrix} a & b \end{bmatrix}$*. The commutativity conditions are:*

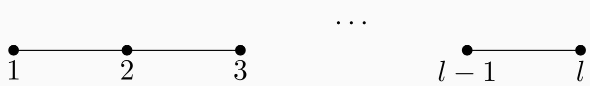
$$\begin{cases}
g_3\phi_\alpha = \psi_\alpha g_1 \implies \begin{bmatrix} a & b \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = c \implies a = c, \\
g_3\phi_\beta = \psi_\beta g_2 \implies \begin{bmatrix} a & b \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = d \implies b = d, \\
g_3\phi_\gamma = \psi_\gamma g_4 \implies \begin{bmatrix} a & b \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = e \implies a + b = e.
\end{cases}$$

The morphism is uniquely determined by the parameters $a, b \in k$*. Hence* $\mathrm{Hom}(M, M'') \cong k^2$*.*

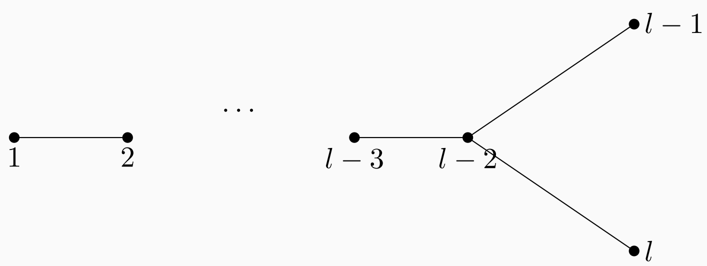
1.4 Dynkin Diagrams and Classical Quivers

Example 1.4.1. *Dynkin Diagrams:*

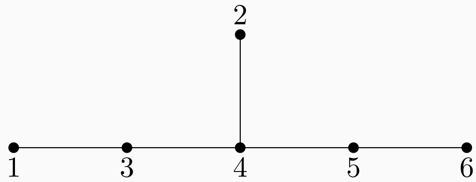
A_l $(l \geq 1)$*:*



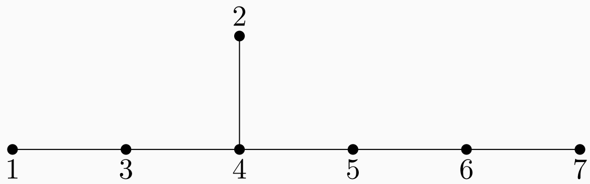
D_l $(l \geq 4)$*:*



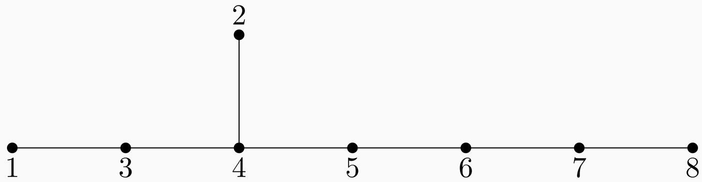
E_6:



E_7:



E_8:



Example 1.4.2. *2-Kronecker quiver:*

$$\begin{tikzcd} 1 \arrow[r, bend left] \arrow[r, bend right] & 2 \end{tikzcd}$$

Representations of this quiver:

$$\begin{tikzcd} V \arrow[r, bend left, "A"] \arrow[r, bend right, "B"'] & W \end{tikzcd}$$

Example 1.4.3. *Jordan quiver:*

$$\begin{tikzcd} 1 \arrow[loop above] \end{tikzcd}$$

Representations of this quiver:

$$\begin{tikzcd} V \arrow[loop above, "A"] \end{tikzcd}$$

$A : V \to V$, the matrix representation of A is similar to a Jordan normal form (when $k = \bar{k}$).

Chapter 2

Lecture 2

Recall the definition:

Definition 2.0.1. *Let $Q = (Q_0, Q_1)$ be a quiver. A representation of Q is a collection*

$$M = (V_i, \varphi_\alpha)_{i \in Q_0, \alpha \in Q_1},$$

where each V_i is a vector space and each arrow $\alpha : i \to j$ is assigned a linear map

$$\varphi_\alpha \in \mathrm{Hom}(V_i, V_j).$$

Example 2.0.2. *Consider the 2-Kronecker quiver*

$$\begin{tikzcd} 1 \arrow[r, bend left] \arrow[r, bend right] & 2 \end{tikzcd}$$

and the representations

$$M: \qquad k \overset{\begin{bmatrix}1\\0\end{bmatrix}}{\underset{\begin{bmatrix}0\\1\end{bmatrix}}{\rightrightarrows}} k^2$$

and

$$N: \qquad k^2 \overset{\begin{bmatrix}1&1\\0&1\end{bmatrix}}{\underset{\begin{bmatrix}1&1\\1&1\end{bmatrix}}{\rightrightarrows}} k^2$$

Denote $M = (V_i, \varphi_\alpha)_{i \in Q_0, \alpha \in Q_1}$ and $N = (U_i, \psi_\alpha)_{i \in Q_0, \alpha \in Q_1}$. Consider the morphism $f : M \to N$. $f = (f_i)_{i \in Q_0}$. $f_i : V_i \to U_i$ and for each $\alpha : i \to j$ in Q, the following *diagram commutes:*

$$\begin{tikzcd} V_i \arrow[r, "\varphi_\alpha"] \arrow[d, "f_i"'] & V_j \arrow[d, "f_j"] \\ U_i \arrow[r, "\psi_\alpha"'] & U_j \end{tikzcd}$$

We use $\mathrm{Rep}(Q)$ *to denote the category of representations of* Q*. For our example, we have*

$$f : M \to N : f = (f_1, f_2) :$$

Let $f = (f_1, f_2) : M \to N$ *be a morphism. Write*

$$f_1 = \begin{bmatrix} a \\ b \end{bmatrix}, \qquad f_2 = \begin{bmatrix} c & d \\ e & f \end{bmatrix}.$$

Then the following diagram commutes:

$$\begin{tikzcd} k \arrow[r, shift left, "\begin{bmatrix} 1 \\ 0 \end{bmatrix}"] \arrow[r, shift right, "\begin{bmatrix} 0 \\ 1 \end{bmatrix}"'] \arrow[d, "f_1=\begin{bmatrix} a \\ b \end{bmatrix}"'] & k^2 \arrow[d, "f_2=\begin{bmatrix} c & d \\ e & f \end{bmatrix}"] \\ k^2 \arrow[r, shift left, "\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}"] \arrow[r, shift right, "\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}"'] & k^2 \end{tikzcd}$$

Commutativity along the two arrows gives

$$f_2 \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} c \\ e \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} f_1 = \begin{bmatrix} a+b \\ b \end{bmatrix},$$

and

$$f_2 \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} d \\ f \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} f_1 = \begin{bmatrix} a+b \\ a+b \end{bmatrix}.$$

Hence

$$c = d = f = a+b, \qquad e = b,$$

so

$$f = \left(\begin{bmatrix} a \\ b \end{bmatrix}, \begin{bmatrix} a+b & a+b \\ b & a+b \end{bmatrix} \right), \qquad \dim \mathrm{Hom}(M, N) = 2.$$

This illustrates how morphisms in $\mathrm{Rep}(Q)$ *are determined by commutative diagrams.*

Definition 2.0.3. *If $M = (V_i, \varphi_\alpha)$ and $N = (U_i, \psi_\alpha)$ are representations of Q, then their direct sum is*

$$M \oplus N = \left(V_i \oplus U_i, \begin{bmatrix} \varphi_\alpha & 0 \\ 0 & \psi_\alpha \end{bmatrix} \right).$$

Inductively, one may form finite direct sums $M_1 \oplus \cdots \oplus M_r$.

Definition 2.0.4. *A representation M is called indecomposable if it cannot be written as a direct sum of two non-zero representations of Q.*

Remark 2.0.5. *For quiver representations, indecomposable and irreducible are different notions; they do not have to coincide.*

Example 2.0.6. *Consider the quiver*

$$\begin{tikzcd} 1 \arrow[r] & 2 & 3 \arrow[l] \end{tikzcd}$$

and the following representations:

$$\begin{tikzcd} M: & k \arrow[r, "1"] & k & 0 \arrow[l, "0"'] \end{tikzcd}$$

$$\begin{tikzcd} N: & k^2 \arrow[r, "\begin{bmatrix} 0 \\ 1 \end{bmatrix}"] & k^2 & k \arrow[l, "\begin{bmatrix} 1 \\ 1 \end{bmatrix}"'] \end{tikzcd}$$

Then

$$\begin{tikzcd} M \oplus N: & k \oplus k^2 \arrow[r, "\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}"] & k \oplus k^2 & 0 \oplus k \arrow[l, "\begin{bmatrix} 0 & 0 \\ 0 & 1 \\ 0 & 1 \end{bmatrix}"'] \end{tikzcd}$$

Indeed, identifying

$$k \oplus k^2 \cong k^3, \qquad k \oplus k^2 \cong k^3, \qquad 0 \oplus k \cong k,$$

we may rewrite this as

$$\begin{tikzcd} k^3 \arrow[r, "\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}"] & k^3 & k \arrow[l, "\begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}"'] \end{tikzcd}$$

Example 2.0.7. *Consider the Jordan quiver:*

$$\begin{tikzcd} \bullet \arrow[loop above] \end{tikzcd}$$

Let A, B, C be three $n \times n$ matrices. Where C denotes an isomorphism (C is invertible). We have the following diagram:

$$\begin{tikzcd} \overset{\bullet}{V} \arrow[loop above, "A"] \arrow[d, "C"'] \\ \overset{\bullet}{V} \arrow[loop below, "B"] \end{tikzcd} \qquad \begin{tikzcd} V \arrow[r, "A"] \arrow[d, "C"'] & V \arrow[d, "C"] \\ V \arrow[r, "B"'] & V \end{tikzcd}$$

Which shows that A and B are similar: $A = C^{-1}BC$. Hence the isomorphism classes are controlled by Jordan normal form, and the indecomposable representations correspond exactly to single Jordan blocks.

To describe $\operatorname{Rep} Q$ we will need to describe all isomorphism classes of indecomposable representations.

Theorem 2.0.8 (Krull–Schmidt)**.** *Any finite representation $M \in \operatorname{Rep} Q$ can be decomposed into finite number of indecomposable representations and its decomposition is unique up to order.*

2.1 Kernel and Cokernel

Recall in linear algebra, we have the following definitions:

Definition 2.1.1. *Let $f : V \to W$ be a linear map. The kernel of f is the set of all vectors $v \in V$ such that $f(v) = 0$.*

Definition 2.1.2. *Let $f : V \to W$ be a linear map. The image of f is the set of all vectors $w \in W$ such that $f(v) = w$ for some $v \in V$.*

Definition 2.1.3. *Let $f : V \to W$ be a linear map. The cokernel of f is the quotient space $W/\operatorname{Im} f$.*

Let $f : M \to N$ ba a morphism from M to N where $M, N \in \operatorname{Rep} Q$. $M = (V_i, \varphi_\alpha)$ and $N = (U_i, \psi_\alpha)$. Define $\ker f \in \operatorname{Rep} Q$ as (L_i, χ_α) where $L_i = \ker f_i$ and $\chi_\alpha = \varphi_\alpha|_{L_i}$. $\forall v_i \in L_i$, $\chi_\alpha(v_i) = \varphi_\alpha(v_i)$.

We want to show that $\ker f$ is a representation in $\operatorname{Rep} Q$. i.e. $\chi_\alpha : L_i \to L_j$ for $\alpha : i \to j$

In fact, $\forall v_i \in L_i$, $f_i(v_i) = 0$, then $f_j(\chi_\alpha(v_i)) = f_j(\varphi_\alpha(v_i)) = \psi_\alpha(f_i(v_i)) = \psi_\alpha(0) = 0$. Hence $\chi_\alpha(v_i) \in \ker f_j = L_j$. Hence $\chi_\alpha : L_i \to L_j$ for $\alpha : i \to j$.

The inclusion $L_i \hookrightarrow V_i$ defines an injective morphism from $\ker f$ to M as the following diagram commutes:

$$\begin{tikzcd} L_i \arrow[r, "\chi_\alpha"] \arrow[d, "incl."] & L_j \arrow[d, "incl."] \\ V_i \arrow[r, "\varphi_\alpha"] & V_j \end{tikzcd}$$

Define $\operatorname{coker} f \in \operatorname{Rep} Q$ as (R_i, ν_α) where $R_i = N_i / \operatorname{Im} f_i$ and $\nu_\alpha : R_i \to R_j$ for $\alpha : i \to j$ is defined as $\nu_\alpha(n_i + f_i(V_i)) = \psi_\alpha(n_i) + f_j(V_j)$.

We need to show that all ν_α are well-defined.

Let $n_i + f_i(V_i) = n_i' + f_i(V_i)$, then $n_i - n_i' \in f_i(V_i)$. That is, $\exists v_i \in V_i$ such that $n_i - n_i' = f_i(v_i)$. Then

$$\psi_\alpha(n_i - n_i') = \psi_\alpha(f_i(v_i)) = f_j(\varphi_\alpha(v_i)) \in f_j(V_j)$$

Hence, $\psi_\alpha(n_i) - \psi_\alpha(n_i') \in f_j(V_j)$. That is, $\nu_\alpha(n_i + f_i(V_i)) = \psi_\alpha(n_i) + f_j(V_j)$ is well-defined.

The projection $N \to \operatorname{coker} f$ is a surjective morphism in $\operatorname{Rep} Q$ as the following diagram commutes:

$$\begin{tikzcd} N_i \arrow[r, "\psi_\alpha"] \arrow[d, "proj."] & N_j \arrow[d, "proj."] \\ R_i \arrow[r, "\nu_\alpha"] & R_j \end{tikzcd}$$

Chapter 3

Lecture 3

Definition 3.0.1. *Recall that for a morophism $f : M \to N$, we have a representation*

$$\ker f \xhookrightarrow{incl.} M$$

A subrepresentation of M is representation L such that $f : L \to M$ is a monomorphism.

Example 3.0.2. *For the quiver* $1 \longrightarrow 2$ *We have trivial representation*

$$0 \xrightarrow{0} 0$$

The following 3 representations are non-trivial

$$k \xrightarrow{0} 0 \quad 0 \xrightarrow{0} k \quad k \xrightarrow{1} k$$

Denoted by S_1, S_2, M.
First two representations are irriducible and indecomposable, the last one is reducible but indecomposable.
To check it, first it is obvious that if M has non-trivial subrepresentations, the only possible ones are S_1 and S_2. For S_1 consider the following diagram:

$$\begin{tikzcd} k \arrow[r, "0"] \arrow[d, "a"'] & 0 \arrow[d, "b"] \\ k \arrow[r, "1"] & k \end{tikzcd}$$

Then a=b=0, it is impossible to construct inclusion.
But if we consider S_2:

$$\begin{tikzcd} 0 \arrow[r, "0"] \arrow[d, "a"'] & k \arrow[d, "b"] \\ k \arrow[r, "1"] & k \end{tikzcd}$$

Then $a = b \cdot 0 = 0$ and b is arbitrary (then we have a monomorphism). Hence we

come to the conclusion that $\mathrm{Hom}(S_1, M) = 0$ *and* $\mathrm{Hom}(S_2, M) = k$*. In other words* S_2 *is a subrepresentation of* M *and hence* M *is reducible.*
But if S_2 *is a direct summand of* M*, then since* S_1 *is not a submodule, we can not have* $M = S_1 \oplus S_2$*. Hence* M *is indecomposable.*

Definition 3.0.3 (Universal property of the kernel)**.** *Let* $f : M \to N$ *be a morphism of representations. A kernel of* f *is a morphism* $g : L \to M$ *such that* $f \circ g = 0$ *and such that, for every morphism* $h : X \to M$ *with* $f \circ h = 0$*, there exists a unique morphism* $u : X \to L$ *making the diagram commute:*

$$\begin{tikzcd} X \arrow[r, dashed, "u"] \arrow[dr, "h"'] & L \arrow[d, "g"] & \\ & M \arrow[r, "f"] & N. \end{tikzcd}$$

Let us show that previous definition $\ker f = (\ker f_i = L_i, \varphi_\alpha|_{L_i})$ is equivalent to the universal property definition with $L = \ker f, g_i = incl.$ i.e. $\ker f_i \xhookrightarrow{incl.} M_i$ where $f_i g_i = 0$.

Proof. Let

$$f = (f_i) : M = (M_i, \varphi_\alpha) \longrightarrow N = (N_i, \rho_\alpha)$$

be a morphism of representations. For each vertex i, set

$$L_i := \ker f_i \subseteq M_i.$$

For each arrow $\alpha : i \to j$, define

$$\psi_\alpha := \varphi_\alpha|_{L_i} : L_i \to L_j.$$

This is well-defined, because if $x \in L_i$, then $f_i(x) = 0$, and since f is a morphism of representations, we have

$$f_j(\varphi_\alpha(x)) = \rho_\alpha(f_i(x)) = \rho_\alpha(0) = 0.$$

Hence $\varphi_\alpha(x) \in \ker f_j = L_j$.

Therefore

$$L = (L_i, \psi_\alpha)$$

is a representation. Let

$$g = (g_i) : L \to M$$

be given by the inclusion maps

$$g_i : L_i = \ker f_i \hookrightarrow M_i.$$

Then for each vertex i,

$$f_i \circ g_i = 0,$$

so

$$f \circ g = 0.$$

We now show that (L, g) satisfies the universal property of the kernel.

Let

$$h = (h_i) : X = (X_i, \nu_\alpha) \to M$$

be any morphism of representations such that

$$f \circ h = 0.$$

Then for every vertex i,

$$f_i \circ h_i = 0.$$

Thus, for every $x_i \in X_i$,

$$f_i(h_i(x_i)) = 0,$$

which means that

$$h_i(x_i) \in \ker f_i = L_i.$$

Hence each h_i factors through L_i, so we may define a linear map

$$u_i : X_i \to L_i, \qquad u_i(x_i) := h_i(x_i).$$

By construction,

$$g_i \circ u_i = h_i$$

for every vertex i.

It remains to check that $u = (u_i) : X \to L$ is a morphism of representations. Let $\alpha : i \to j$ be an arrow and let $x_i \in X_i$. Then

$$\psi_\alpha(u_i(x_i)) = \psi_\alpha(h_i(x_i)) = \varphi_\alpha(h_i(x_i)) = h_j(\nu_\alpha(x_i)) = u_j(\nu_\alpha(x_i)).$$

Therefore

$$\psi_\alpha \circ u_i = u_j \circ \nu_\alpha$$

for every arrow α, so u is indeed a morphism of representations.

Finally, uniqueness is clear. If

$$u' = (u'_i) : X \to L$$

is another morphism such that

$$g \circ u' = h,$$

then for each vertex i,

$$g_i \circ u'_i = h_i.$$

Since $g_i : L_i \hookrightarrow M_i$ is just the inclusion map, it follows that

$$u_i'(x_i) = h_i(x_i) = u_i(x_i)$$

for all $x_i \in X_i$. Hence $u_i' = u_i$ for all i, and so $u' = u$.

Thus (L, g) satisfies the universal property of the kernel. Therefore the representation

$$L = (\ker f_i, \varphi_\alpha|_{\ker f_i})$$

with the inclusion morphism $g : L \to M$ is precisely the kernel of f. $\square$

Analogously we have equivalent definition of $\operatorname{coker} f$.

Definition 3.0.4 (Cokernel of a morphism)**.** *Let*

$$f = (f_i) : M = (M_i, \varphi_\alpha) \to N = (N_i, \psi_\alpha)$$

be a morphism of representations.
For each vertex $i \in Q_0$, set

$$H_i := N_i / f_i(M_i).$$

For each arrow $\alpha : i \to j$, define

$$\chi_\alpha : H_i \to H_j, \qquad \chi_\alpha\big(n_i + f_i(M_i)\big) := \psi_\alpha(n_i) + f_j(M_j).$$

Then

$$\operatorname{coker} f := (H_i, \chi_\alpha).$$

Equivalently, a cokernel of f is a morphism

$$g : N \to H$$

such that $g \circ f = 0$ and satisfying the following universal property:
for every representation Y and every morphism $h : N \to Y$ with $h \circ f = 0$, there exists a unique morphism $u : H \to Y$ such that

$$u \circ g = h.$$

Equivalently, the following diagram commutes:

$$\begin{tikzcd} M \arrow[r, "f"] & N \arrow[r, "g"] \arrow[dr, "h"'] & H \arrow[d, dashed, "u"] \\ & & Y \end{tikzcd}$$

Exercise 3.0.5. *Prove the quivalence of the two definitions of* $\operatorname{coker} f$.

Proof. Let

$$f = (f_i) : M = (M_i, \varphi_\alpha) \to N = (N_i, \psi_\alpha)$$

be a morphism of representations. For each vertex i, define

$$H_i := N_i/f_i(M_i).$$

For each arrow $\alpha : i \to j$, define

$$\chi_\alpha\big(n_i + f_i(M_i)\big) := \psi_\alpha(n_i) + f_j(M_j).$$

We first check that χ_α is well-defined. Suppose

$$n_i + f_i(M_i) = n_i' + f_i(M_i).$$

Then

$$n_i - n_i' = f_i(m_i)$$

for some $m_i \in M_i$. Hence

$$\psi_\alpha(n_i) - \psi_\alpha(n_i') = \psi_\alpha(f_i(m_i)).$$

Since f is a morphism of representations, we have

$$\psi_\alpha \circ f_i = f_j \circ \varphi_\alpha.$$

Therefore

$$\psi_\alpha(f_i(m_i)) = f_j(\varphi_\alpha(m_i)) \in f_j(M_j).$$

So

$$\psi_\alpha(n_i) + f_j(M_j) = \psi_\alpha(n_i') + f_j(M_j),$$

and thus χ_α is well-defined.

Hence

$$H = (H_i, \chi_\alpha)$$

is a representation.

Now let

$$g = (g_i) : N \to H$$

be the natural quotient morphism, where

$$g_i : N_i \to N_i/f_i(M_i), \qquad g_i(n_i) = n_i + f_i(M_i).$$

Then for every vertex i and every $m_i \in M_i$,

$$(g_i \circ f_i)(m_i) = f_i(m_i) + f_i(M_i) = 0.$$

Thus

$$g \circ f = 0.$$

We now prove the universal property. Let

$$h = (h_i) : N \to Y = (Y_i, \eta_\alpha)$$

be a morphism such that

$$h \circ f = 0.$$

Then for each vertex i,

$$h_i \circ f_i = 0.$$

Hence h_i vanishes on $f_i(M_i)$, so it factors uniquely through the quotient $N_i/f_i(M_i)$. Therefore there exists a unique linear map

$$u_i : H_i = N_i/f_i(M_i) \to Y_i$$

such that

$$u_i(n_i + f_i(M_i)) := h_i(n_i).$$

This is well-defined because if

$$n_i + f_i(M_i) = n'_i + f_i(M_i),$$

then $n_i - n'_i = f_i(m_i)$ for some $m_i \in M_i$, and hence

$$h_i(n_i) - h_i(n'_i) = h_i(f_i(m_i)) = 0.$$

By construction,

$$u_i \circ g_i = h_i$$

for every vertex i.

It remains to check that $u = (u_i) : H \to Y$ is a morphism of representations. Let $\alpha : i \to j$ be an arrow and let $n_i + f_i(M_i) \in H_i$. Then

$$\eta_\alpha\big(u_i(n_i+f_i(M_i))\big) = \eta_\alpha(h_i(n_i)) = h_j(\psi_\alpha(n_i)) = u_j\big(\psi_\alpha(n_i)+f_j(M_j)\big) = u_j\big(\chi_\alpha(n_i+f_i(M_i))\big).$$

Therefore

$$\eta_\alpha \circ u_i = u_j \circ \chi_\alpha,$$

so u is a morphism of representations.

Finally, uniqueness is clear: if

$$u' = (u'_i) : H \to Y$$

also satisfies

$$u' \circ g = h,$$

then for each i,

$$u_i'(n_i + f_i(M_i)) = u_i'(g_i(n_i)) = h_i(n_i) = u_i(n_i + f_i(M_i)).$$

Thus $u_i' = u_i$ for all i, so $u' = u$.

Therefore the quotient representation

$$\big(N_i/f_i(M_i), \chi_\alpha\big)$$

with the natural projection $g : N \to H$ satisfies the universal property of the cokernel. $\square$

Proof. Suppose

$$g = (g_i) : N = (N_i, \psi_\alpha) \to H = (H_i, \chi_\alpha)$$

is a cokernel of

$$f = (f_i) : M = (M_i, \varphi_\alpha) \to N = (N_i, \psi_\alpha)$$

in the sense of the universal property.

Let

$$C := \big(N_i/f_i(M_i),\ \overline{\psi}_\alpha\big)$$

be the pointwise cokernel representation, where

$$\overline{\psi}_\alpha(n_i + f_i(M_i)) := \psi_\alpha(n_i) + f_j(M_j).$$

Let

$$\pi = (\pi_i) : N \to C$$

be the natural quotient morphism. As shown above,

$$\pi \circ f = 0.$$

Since $g : N \to H$ is a cokernel of f, applying its universal property to the morphism

$$\pi : N \to C$$

with $\pi \circ f = 0$, there exists a unique morphism

$$a : H \to C$$

such that

$$a \circ g = \pi.$$

On the other hand, since $g \circ f = 0$, for each vertex i we have

$$g_i \circ f_i = 0.$$

Thus g_i vanishes on $f_i(M_i)$, so it factors uniquely through the quotient

$$N_i/f_i(M_i).$$

Hence for each i there exists a unique linear map

$$b_i : C_i = N_i/f_i(M_i) \to H_i$$

given by

$$b_i(n_i + f_i(M_i)) := g_i(n_i).$$

These maps assemble into a morphism

$$b : C \to H,$$

because for any arrow $\alpha : i \to j$,

$$\chi_\alpha\big(b_i(n_i+f_i(M_i))\big) = \chi_\alpha(g_i(n_i)) = g_j(\psi_\alpha(n_i)) = b_j\big(\psi_\alpha(n_i)+f_j(M_j)\big) = b_j\big(\overline{\psi}_\alpha(n_i+f_i(M_i))\big).$$

So

$$b \circ \pi = g.$$

We now show that a and b are inverse isomorphisms.

First,

$$(a \circ b) \circ \pi = a \circ (b \circ \pi) = a \circ g = \pi.$$

Also,

$$\mathrm{id}_C \circ \pi = \pi.$$

Since $\pi : N \to C$ is a cokernel of f, the uniqueness part of its universal property implies

$$a \circ b = \mathrm{id}_C .$$

Similarly,

$$(b \circ a) \circ g = b \circ (a \circ g) = b \circ \pi = g,$$

and also

$$\mathrm{id}_H \circ g = g.$$

Since $g : N \to H$ is a cokernel of f, the uniqueness part of its universal property implies

$$b \circ a = \mathrm{id}_H .$$

Therefore a and b are inverse isomorphisms, and hence

$$H \cong C = \big(N_i/f_i(M_i),\ \overline{\psi}_\alpha\big).$$

In particular, for each vertex i we have

$$H_i \cong N_i/f_i(M_i),$$

and under these identifications, for every arrow $\alpha : i \to j$,

$$\chi_\alpha = a_j^{-1} \circ \overline{\psi}_\alpha \circ a_i.$$

Thus any cokernel given by the universal property is isomorphic to the pointwise quotient representation

$$\big(N_i/f_i(M_i),\ \psi_\alpha \text{ induced on the quotients}\big).$$

$\square$

Theorem 3.0.6 (the first isomorphism theorem)**.** *If $f : M \to N$ is a morphism of representations, then* $\operatorname{im} f \cong M/\ker f$*.*

Proof. Write

$$M = (M_i, \varphi_\alpha), \qquad N = (N_i, \psi_\alpha).$$

Then

$$\operatorname{im} f = (\operatorname{im} f_i, \psi_\alpha|_{\operatorname{im} f_i}).$$

This is a representation, since for every arrow $\alpha : i \to j$ and every $m_i \in M_i$,

$$\psi_\alpha(f_i(m_i)) = f_j(\varphi_\alpha(m_i)) \in \operatorname{im} f_j.$$

Also,

$$M/\ker f = \big(M_i/\ker f_i, \overline{\varphi}_\alpha\big),$$

where for $\alpha : i \to j$,

$$\overline{\varphi}_\alpha(m_i + \ker f_i) = \varphi_\alpha(m_i) + \ker f_j.$$

For each vertex i, the first isomorphism theorem for vector spaces gives an isomorphism

$$\overline{f}_i : M_i/\ker f_i \longrightarrow \operatorname{im} f_i, \qquad \overline{f}_i(m_i + \ker f_i) = f_i(m_i).$$

We claim that $\overline{f} = (\overline{f}_i)$ is a morphism of representations. Consider the diagram

$$\begin{tikzcd}
M_i/\ker f_i \arrow[r, "\overline{\varphi}_\alpha"] \arrow[d, "\overline{f}_i"'] & M_j/\ker f_j \arrow[d, "\overline{f}_j"] \\
\operatorname{im} f_i \arrow[r, "\psi_\alpha|_{\operatorname{im} f_i}"] & \operatorname{im} f_j
\end{tikzcd}$$

For any $m_i \in M_i$,

$$\overline{f}_j\big(\overline{\varphi}_\alpha(m_i+\ker f_i)\big) = \overline{f}_j\big(\varphi_\alpha(m_i)+\ker f_j\big) = f_j(\varphi_\alpha(m_i)) = \psi_\alpha(f_i(m_i)) = \psi_\alpha|_{\operatorname{im} f_i}\big(\overline{f}_i(m_i+\ker f_i)\big).$$

Hence the diagram commutes for every arrow α, so $\overline{f} : M/\ker f \to \operatorname{im} f$ is a morphism in $\operatorname{Rep} Q$. Since each $\overline{f}_i$ is an isomorphism, $\overline{f}$ is an isomorphism of representations. Therefore

$$M/\ker f \cong \operatorname{im} f.$$

$\square$

These all together shows that $\operatorname{Rep} Q$ is abelian category. More precisely,

Definition 3.0.7. *We call a category $\mathcal{C}$ abelian k-category if the following conditions are satisfied:*

1. *$\mathcal{C}$ is k-category, for any $X, Y \in \operatorname{Ob}(\mathcal{C})$, $\operatorname{Hom}_{\mathcal{C}}(X, Y)$ is a vector space over k.*
2. *$\mathcal{C}$ is a additive category, $\mathcal{C}$ has direct sums and zero object.*
3. *Each morphism $f : X \to Y$ has kernel K and cokernel H. $f : M \to N$, $i : K \to M$, $p : N \to H$. and $\operatorname{coker} i \cong \ker p$.*

Definition 3.0.8. *Let* $M \longrightarrow N \longrightarrow P$ *be a sequence of morphisms in* $\operatorname{Rep} Q$. *We say that this sequence is exact if* $\ker p = \operatorname{im} f$.

Definition 3.0.9. *We call a sequence of morphisms in* $\operatorname{Rep} Q$ *a short exact sequence if it is exact and of the form*

$$0 \longrightarrow L \xrightarrow{f} M \xrightarrow{g} N \longrightarrow 0$$

Where f is injective, g is surjective $\operatorname{im} f = \ker g$

For $f : M \to N$ we have exact sequence:

$$0 \longrightarrow \ker f \xrightarrow{i} M \xrightarrow{f} N \xrightarrow{p} \operatorname{coker} f \longrightarrow 0$$

Where i is an inclusion and p is a projection.

Or we have short exact sequence:

$$0 \longrightarrow \ker f \xrightarrow{i} M \xrightarrow{p} M/\ker f \longrightarrow 0$$

Example 3.0.10. ***Quiver and Basic Representations:*** *The quiver is given by:*

$$1 \longrightarrow 2$$

Consider the following representations:

- S_1*:* $k \xrightarrow{0} 0$

- S_2: $0 \xrightarrow{0} k$
- M: $k \xrightarrow{1} k$

Exact Sequences: *The following short exact sequences are presented:*

$$0 \longrightarrow S_2 \xrightarrow{\text{incl}\,(0,1)} M \xrightarrow{\ker f\,(1,0)} S_1 \longrightarrow 0$$

$$0 \longrightarrow S_2 \xrightarrow{(0,1)} S_1 \oplus S_2 \xrightarrow{(1,0)} S_1 \longrightarrow 0$$

Both sequences are exact.

Definition 3.0.11. *A morphism $f : L \to M$ is a section if there exists a morphism $h : M \to L$ such that $hf = 1_L$. A morphism $g : M \to N$ is a retraction if there exists a morphism $h : N \to M$ such that $gh = 1_N$.*

Definition 3.0.12. *We say that a short exact sequence $0 \longrightarrow L \xrightarrow{f} M \xrightarrow{g} N \longrightarrow 0$ splits if f has a retraction or equivalently f is a section.*

Theorem 3.0.13. *Let*

$$0 \longrightarrow L \xrightarrow{f} M \xrightarrow{g} N \longrightarrow 0$$

be a short exact sequence in $\operatorname{Rep} Q$. *Then*

1. *f is a section if and only if g is a retraction.*
2. *If f is a seciton then $\ker g = \operatorname{im} f$ is a direct summand of M.*

Proof. $(\Rightarrow)$: Suppose f is a section. $\exists h : M \to L$ such that $hf = 1_L$. We will define $h' : N \to M$ and show that $gh' = 1_N$. $\forall n \in N$, since g is surjective $\exists m$ such that $n = g(m)$. $h'(n) = m - f(h(m))$ where $h' : N \to M$. We have to show that h' is well-defined.

Suppose m, m' such that $n = g(m) = g(m')$. We have to show that $m' - f(h(m')) = m - f(h(m))$. $m' - m = f(h(m')) - f(h(m)) \in \operatorname{im} f$, $\exists l \in L$ such that $f(l) = m' - m =$. Now:

$$m' - f(h(m')) - m + f(h(m)) = m' - m - f(h(m' - m)) = m' - m - f(h(f(l))) = m' - m - f(l) = 0$$

$\square$

Chapter 4

Lecture 4

Recall short exact sequence:

$$0 \longrightarrow L \xrightarrow{f} M \xrightarrow{g} N \longrightarrow 0$$

Where $\ker f = L$, $\operatorname{im} f = \ker g$, $\operatorname{im} g = N$.

f is section if $\exists h : M \to L$ such that $hf = 1_L$. g is retraction if $\exists h' : N \to M$ such that $gh' = 1_N$.

Theorem 4.0.1. *Taking a short exact sequence*

$$0 \longrightarrow L \xrightarrow{f} M \xrightarrow{g} N \longrightarrow 0$$

1. *f is a section if and only if g is a retraction.*
2. *If f is a seciton then $\ker g = \operatorname{im} f$ is a direct summand of M.*

proof of (a). ($\Rightarrow$): Suppose f is a section. $\exists h : M \to L$ such that $hf = 1_L$. We will define $h' : N \to M$ and show that $gh' = 1_N$. $\forall n \in N$, since g is surjective $\exists m$ such that $n = g(m)$. $h'(n) = m - f(h(m))$ where $h' : N \to M$. We have to show that h' is well-defined. Suppose m, m' such that $n = g(m) = g(m')$. We have to show that $m' - f(h(m')) = m - f(h(m))$. $m' - m = f(h(m')) - f(h(m)) \in \operatorname{im} f$, $\exists l \in L$ such that $f(l) = m' - m =$. Now:

We need to show h' is a morphism. Let $M = (M_i, \varphi_\alpha), L = (L_i, \varphi'_\alpha), N = (N_i, \varphi''_\alpha)$. Then f, g, h are morphisms of representations. We have to show that h' is a morphism of representations. Let $\alpha : i \to j$ be an arrow in Q. Then consider the following diagram:

$$\begin{tikzcd}
\underset{\bullet}{L_i} \arrow[r, "\varphi'_\alpha"] \arrow[d, shift right, "f_i"'] & \underset{\bullet}{L_j} \arrow[d, shift left, "f_j"] \\
\underset{\bullet}{M_i} \arrow[r, "\varphi_\alpha"] \arrow[u, shift right, "h_i"'] \arrow[d, shift right, "g_i"'] & \underset{\bullet}{M_j} \arrow[u, shift left, "h_j"] \arrow[d, shift left, "g_j"] \\
\underset{\bullet}{N_i} \arrow[r, "\varphi''_\alpha"] \arrow[u, shift right, "h'_i"'] & \underset{\bullet}{N_j} \arrow[u, shift left, "h'_j"]
\end{tikzcd}$$

Take $n_i \in N_i$, and for n_i we choose m_i such that $g_i(m_i) = n_i, h'_i(n_i) = m_i - f_i(h_i(m_i))$. Then we have

$$\begin{aligned}
\varphi_\alpha h'_i(n_i) &= \varphi_\alpha(m_i - f_i(h_i(m_i))) \\
&= \varphi_\alpha(m_i) - \varphi_\alpha(f_i(h_i(m_i))) \\
&= \varphi_\alpha(m_i) - f_j(\psi'_\alpha(h_i(m_i))) \\
&= \varphi_\alpha(m_i) - f_j(h_j(\varphi_\alpha(m_i)))
\end{aligned}$$

And

$$\begin{aligned}
h'_j(\varphi''_\alpha(n_i)) &= h'_j(\varphi''_\alpha(g_i(m_i))) \\
&= h'_j(g_j(\varphi_\alpha(m_i))) \\
&= \varphi_\alpha(m_i) - f_j(h_j(\varphi_\alpha(m_i)))
\end{aligned}$$

Hence h' is a morphism of representations.

The only thing need to show now is that $gh' = 1_N$. Let $n_i \in N_i$, then $\exists m_i \in M_i$ such that $n_i = g(m_i)$. Then $g_i h'_i(n_i) = g_i(m - fh(m)) = g(m) - g(fh(m)) = g_i(m_i) = n_i$. Hence $gh' = 1_N$.

($\Rightarrow$): Suppose g is a retraction. $\exists h' : N \to M$ such that $gh' = 1_N$. We will define $h : M \to L$ and show that $hf = 1_L$.

For any m we have $g(m - h'g(m)) = g(m) - g(h'g(m)) = g(m) - g(m) = 0$. Hence $m - h'g(m) \in \ker g = \operatorname{im} f$. Hence $\exists! l \in L$ such that $f(l) = m - h'g(m)$ since f is injective. Define $h(m) = l$. Then

We have to show that h is a morphism of representations. Let $M = (M_i, \varphi_\alpha), L = (L_i, \varphi'_\alpha), N = (N_i, \varphi''_\alpha)$. Then f, g, h' are morphisms of representations. We have to show that h is a morphism of representations. Let $\alpha : i \to j$ be an arrow in Q. Then consider the following diagram:

$$\begin{tikzcd}
\overset{L_i}{\bullet} \arrow[r, "\varphi'_\alpha"] \arrow[d, shift right, "f_i"'] & \overset{L_j}{\bullet} \arrow[d, shift left, "f_j"] \\
\overset{M_i}{\bullet} \arrow[r, "\varphi_\alpha"] \arrow[u, shift right, "h_i"'] \arrow[d, shift right, "g_i"'] & \overset{M_j}{\bullet} \arrow[u, shift left, "h_j"] \arrow[d, shift left, "g_j"] \\
\overset{N_i}{\bullet} \arrow[r, "\varphi''_\alpha"] \arrow[u, shift right, "h'_i"'] & \overset{NN_j}{\bullet} \arrow[u, shift left, "h'_j"]
\end{tikzcd}$$

Take $m_i \in M_i$, and for m_i we choose $l_i \in L_i$ such that $f(l_i) = m_i - h'_i g_i(m_i)$ and $h_i(m_i) = l_i$. Then we have

$$\varphi_\alpha(h_i(m_i)) = \varphi'_\alpha(l_i)$$

On the other hand,

$$\begin{aligned}
h_j \varphi_\alpha(m_i) &= h_j \varphi_\alpha(f_i(l_i) + h'_i g_i(m_i)) \\
&=
\end{aligned}$$

□

proof of (b). Suppose f is a section then g is a retraction. $\forall m \in M$, $\exists h : M \to L$ such that $hf = 1_L$ and $\exists h' : N \to M$ such that $gh' = 1_N$. Then $m = h'g(m) + (m - h'g(m))$ where $h'g(m) \in \operatorname{im} h'$ and $m - h'g(m) \in \ker g$. And $\operatorname{im} h' \cap \ker g = 0$. Thus $M_i = \ker g \oplus \operatorname{im} h'$ as vector spaces.

Furthermore we need to show that $\ker g \oplus \operatorname{im} h'$ is a direct sum of representations. i.e.

$$\varpi_\alpha = \begin{bmatrix} \varphi_\alpha|_{\ker g_i} & 0 \\ 0 & \varphi_\alpha|_{\operatorname{im} h'_i} \end{bmatrix}$$

Let $m_i \in \ker g_i$, then $0 = \varphi''_\alpha(g_i(m_i)) = g_j \varphi_\alpha(m_i)$ where $\varphi_\alpha(m_i) \in \ker g_j$ □

Corollary 4.0.2. *If*

$$0 \longrightarrow L \xrightarrow{f} M \xrightarrow{g} N \longrightarrow 0$$

is a short exact split sequence in $\operatorname{Rep} Q$*, then* $M = L \oplus N$ *as representations.*

Proof. f is injective, $L \cong \operatorname{im} f = \ker g$. By isomorphism theorem $M/\ker g \cong N$. Since g is surjective, $N \cong \operatorname{im} g$. Statement follows from (b) of the previous theorem. □

Example 4.0.3. *not split sequence:* S_1, S_2, M.

4.1 Hom Functor

Let $\mathcal{C}$ and $\mathcal{C}'$ be two categories. A **covariant functor** $F : \mathcal{C} \to \mathcal{C}'$ is a map that assigns to each object $X \in \mathcal{C}$ an object $F(X) \in \mathcal{C}'$ and to each morphism $f : X \to Y$ in $\mathcal{C}$ a morphism $F(f) : F(X) \to F(Y)$ in $\mathcal{C}'$, satisfying the following conditions:

- $F(f \circ g) = F(f) \circ F(g)$ for all morphisms $f : X \to Y$ and $g : Y \to Z$ in $\mathcal{C}$.
- $F(\mathrm{id}_X) = \mathrm{id}_{F(X)}$ for all objects $X \in \mathcal{C}$.

If $F(f \circ g) = F(g) \circ F(f)$ for all morphisms $f : X \to Y$ and $g : Y \to Z$ in $\mathcal{C}$, then F is called a **contravariant functor**.

If $X \in \operatorname{Rep} Q$ then $\operatorname{Hom}_{\mathcal{C}}(X, -)$ is a covariant functor from $\operatorname{Rep} Q$ to the category of k-vector spaces.

Example 4.1.1. *Fix* $X \in \operatorname{Rep} Q$*. Let* $M, N \in \operatorname{Rep} Q$*. Then* $\operatorname{Hom}_{\operatorname{Rep} Q}(X, M)$ *is a* k*-vector space. For a morphism* $f : M \to N$*, define*

$$\operatorname{Hom}(X, f) := f_* : \operatorname{Hom}_{\operatorname{Rep} Q}(X, M) \to \operatorname{Hom}_{\operatorname{Rep} Q}(X, N), \qquad f_*(g) = f \circ g.$$

This is a k*-linear map.*